Q.26 Explain the concept of "skin effect"in conductors and its implications for transmission. (CO2)

Q.27 Define string efficiency. How it is improved? (CO2)

Q.28 Explain sag in transmission line. (CO3)

Q.29 Differentiate between short, medium and long transmission line. (CO1)

Q.30 What are the major components of an A.C. distribution system? (CO5)

Q.31 Discuss the impact of environment factors on transmission line design. (CO5)

Q.32 Explain the working principle of circuits breakers in transmission systems. (CO2)

Q.33 Describe the methods used for voltage regulation in distribution systems. (CO3)

Q.34 Discuss the importance of power factor correction in transmission and distribution systems. (CO4)

Q.35 What are the different types of insulators used in transmission lines and their functions? (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Compare and contrast the A.C. and D.C. transmission systems in terms of efficiency, cost, and application. Provide examples where each system is preferable. (CO1)

Q.37 Explain in detail about HVDC system and its features. (CO4)

Q.38 Explain feeder and distributor and what are the factors considered in the design of feeder? (CO5)

No. of Printed Pages : 4 202433

Roll No. ....

3rd Sem / Mechatronics

Subject:- Electric Power Transmission and Distribution

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is NOT a primary function of a transmission line? (CO1)

a) Voltage transformation
b) Power distribution
c) Power transmission
d) Line protection

Q.2 The main purpose of using extra high voltage (EHV) transmission is to: (CO3)

a) Increase losses b) Reduce line size
c) Minimize power loss d) Enhance insulation

Q.3 The inductance of a transmission line primarily affects: (CO4)

a) voltage level
b) power factor
c) Current capacity
d) Insulation requirements

Q.4 Which of the following components is essential for converting AC to DC in a distribution system?(CO3)

a) Transformer b) Rectifier
c) Inverter d) Circuit breaker

Q.5 The main advantage of three-phase AC systems over single-phase system is: (CO3)

a) Higher voltage levels
b) Less current
c) Reduced power losses
d) Balanced load distribution

Q.6 The term "line capacitance" in transmission line refers to: (CO3)
a) Energy storage capability
b) Resistance to current flow
c) Power factor improvement
d) Voltage drop minimization

Q.7 Which of the following is a common material used for overhead transmission conductors? (CO1)
a) Copper b) Steel
c) Aluminum d) All of the above

Q.8 In an A.C. distribution system, the purpose of a transformer is to: (CO2)
a) Increase current
b) Decrease voltage
c) Change voltage levels
d) Generate electricity

Q.9 The 'Ferranti effect' is observed in: (CO3)
a) Long transmission lines
b) Short transmission lines
c) Distribution networks
d) Underground cables

Q.10 What does the term 'line loading' refer to? (CO4)
a) The mechanical stress on the line
b) The electrical load carried by the line
c) The temperature of the line
d) The length of the transmission line

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What is the standards voltage level used in India? (CO1)

Q.12 Long transmission line is used for the length more than_______ (CO1)

Q.13 Step up transformer is used near_______ (CO2)

Q.14 The Resistance of a conductor is primarily affected by its _____ and______. (CO2)

Q.15 The load on a transmission line is often measured in_______. (CO3)

Q.16 In A.C. system, the phase difference between voltage and current is known as the______. (CO4)

Q.17 _____ systems are used to distribute power to residential areas. (CO2)

Q.18 The term______ refers to the energy loss due to heat in conductors. (CO3)

Q.19 _____is a major advantage of using EHV transmission over conventional transmission lines. (CO3)

Q.20 If the supply frequency is increased, the skin effect will_______ (CO2)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the basic principles of electric power transmission and distribution. (CO1)

Q.22 Discuss the significance of line parameters (R,L,C) in transmission line performance (CO2)

Q.23 Describe the factors influencing the choice of transmission voltage levels. (CO3)

Q.24 What are the advantages of using EHV transmission lines? (CO3)

Q.25 How does the length of a transmission line affect its performance? (CO1)